Key Mistakes And Lessons Learnt While Designing And Scaling Real-World IoT And AIoT Products



Prototypes impress and pilots succeed, but scaling to thousands of devices across cities exposes what the lab never tested. Mandar Halbhavi of Jio shares five engineering lessons for building IoT and AIoT products that last in the real world.

An IoT product that performs flawlessly in the laboratory may still fail when deployed across a city or nationwide. The difference lies in how well it has been engineered for scalability, reliability, and long-term serviceability.

Why pilots succeed and scale-ups struggle

Developing a product to solve a specific problem is relatively straightforward. Prototypes usually perform well and pilot deployments are often successful because experienced engineers closely monitor installations, resolve issues immediately, and intervene manually when necessary.

The situation changes once deployments move beyond a handful of installations. As products are rolled out across cities or states, they operate in environments that are far less predictable than laboratory conditions. Network availability varies by location, power quality differs across regions, environmental conditions become harsher, and installation and maintenance teams may have different levels of technical expertise.

A product that performs flawlessly in the laboratory may therefore struggle in the field, not because of poor engineering but because real-world operating conditions were never considered during its design. Scalability depends not only on whether a product functions correctly, but also on whether it can operate reliably under diverse conditions while remaining easy to deploy, maintain, and support.

Building scalability into product design

Scalability is best treated as a design objective rather than a post-deployment requirement. Alongside hardware and software development, engineers need to evaluate how products will perform in different environments, how they will be deployed in the field, and how they will be maintained throughout their operational life.

Several practical factors influence large-scale deployments, including environmental conditions, installation procedures, manufacturing